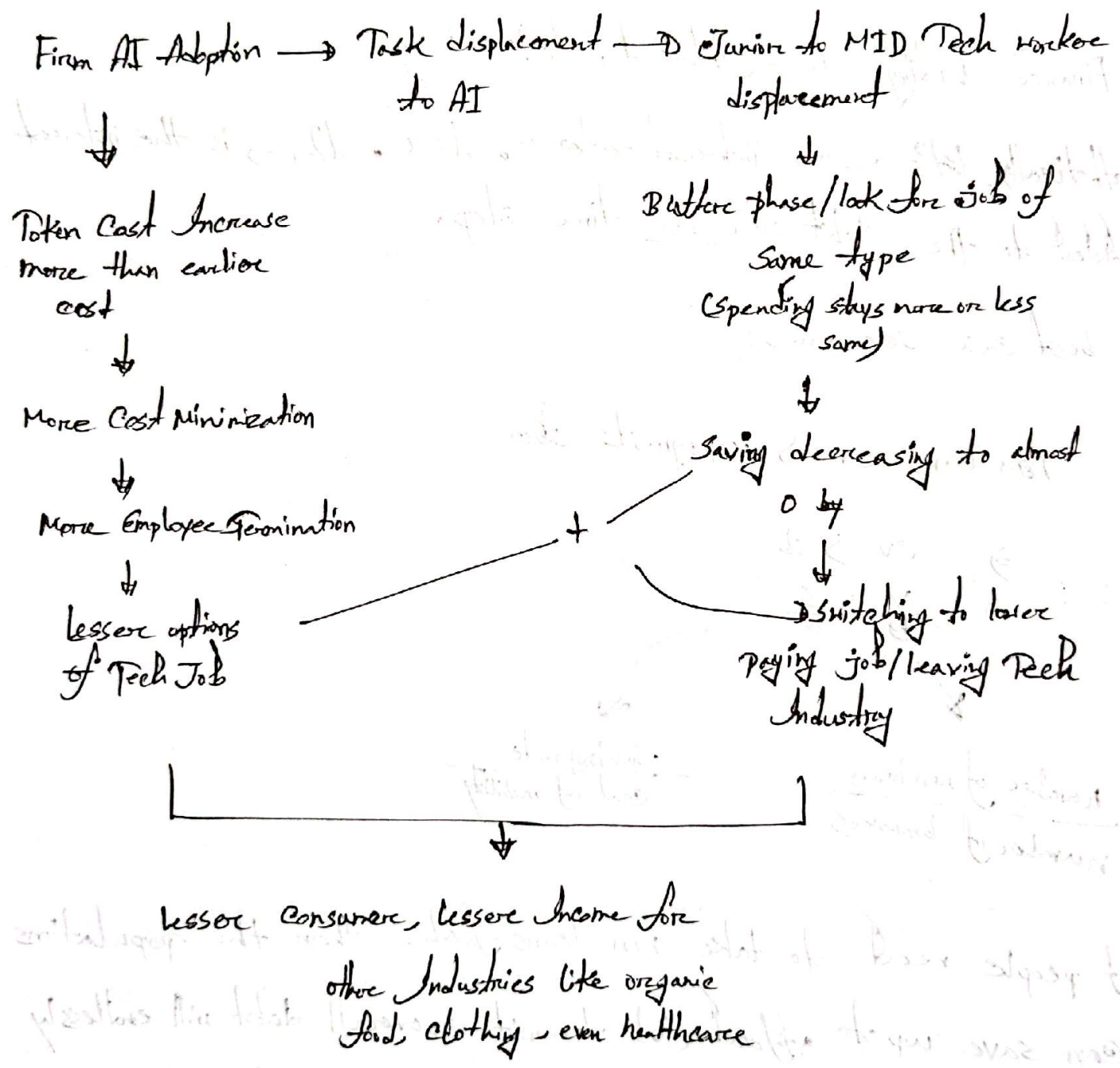


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Let's do the same model for
Tech Industry after AI



Firms are giving task to AI and firing human. So, now one human need to supervise many tasks at once, so their individual workloading is increasing. And AI token price are increasing as well.

$$Y = \left(\sum_{i=1}^N y_i^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}$$

[from Acemoglu and Restrepo 18]

Y is the output task, combining unit task y_i per month

$N-1$ to $N \rightarrow$ band of task the firm needs

$\epsilon \rightarrow$ Elasticity of substitution betⁿ tasks. If task A's cost rises to 1% ($\epsilon > 0$) relative to task B, then ratio of A's quantity to B falls by $\epsilon\%$.

$\epsilon = \begin{cases} 0 \rightarrow \text{fixed proportion, every task needs a set amount} \\ 1 \rightarrow \text{Cobb Douglas} \rightarrow \text{same share of budget on every tasks} \\ > 1 \rightarrow \text{easy to swap} \end{cases}$

Task Production cost: Task i can be performed by both human and AI.

For that, effective unit cost of production is $e(i)$:

$$e(i) = \frac{w}{r(i)} \quad (\text{AR 18})$$

$e(i) \rightarrow$ Effective cost of labor in producing task i

$w \rightarrow$ Market wage for each task monthly.

$r(i) \rightarrow$ Task specific productivity from labor

For AI, for 1 unit of task, require q tokens priced at p_t per token, AI cost per unit of task output c_A is:

$$Q_A = P_A Q$$

$C_A \rightarrow$ unit cost of AI per unit of task output

$P_T \rightarrow$ price of token

$Q \rightarrow$ quantity of AI tokens required to execute one unit of task output.

Assumption here: 1) $\gamma(i)$ continuous and increasing

2) C_A is constant (let's assume tokens are not getting more expensive) $\rightarrow$ (for a year)

Task Allocation: human / AI: Any firm will choose the cheapest factor for each task i , for unit task (Priced $P(i)$)

$$P(i) = \begin{cases} \min\{C_A, \frac{w}{\gamma(i)}\} & \text{if } i \leq I \\ \frac{w}{\gamma(i)} & \text{if } i > I \end{cases}$$

$P(i)$: Minimum unit price/cost of producing task i

I : Technological capability threshold, tasks $i \leq I$ can technically be automated by AI

$\bar{I} \rightarrow$ Threshold task where effective labor cost equals unit AI cost.

$$I^* = \min\{I, \bar{I}\}$$

Actual task threshold executed by AI in equilibrium

$\bar{I}$ occurs when labor cost equal = AI token cost.

$$\frac{w}{r(\bar{I})} = c_A \quad \Rightarrow \quad \frac{w}{r(\bar{I})} = c_A$$

$$\Rightarrow r(\bar{I}) = \frac{w}{c_A}$$

(AR 18 eq (6))

[tasks $i \in [N-1, I^*]$ produced by AI, tasks $i \in (I^*, N]$ are produced by human labor]

Assumptions here:

- if $I \leq \bar{I}$, automation is constrained by Tech
- if $I \geq \bar{I}$, automation is limited by price

Tech Employment and spending allocation:

$$L = \gamma w^{-\frac{1}{\sigma}} \int_{I^*}^N r(i)^{\frac{1}{\sigma}-1} di$$

$\swarrow$ Aggregated Tech employment Demand
 $\searrow$ Wage Rate
 $\nearrow$ Productivity in task i

$$L = \frac{\alpha}{C_A \tau \epsilon_s} \cdot \frac{1}{\beta s}$$

when junior Tech role goes to AI

Wage share out of total firm spending on task $x \Rightarrow$

$$S_u = \frac{wL}{x} \quad (\text{AR 18})$$

Worker Displacement:

Search duration for a job $\tau = t - t_d$
current time $\downarrow$ firing date

Saving of the worker: $\frac{dm(w)}{dw} = z - \bar{c}$
Income per unit time (severance, freelancing, benefits, $z > 0$)
 $\downarrow$ worker's spending rate per unit time

Initial saving $\leftarrow m(0) = s_r W$ Pre fire wage rate
 $\downarrow$ Pre fire saving rate

$$m(w) = s_r W + (z - \bar{c})\tau$$

Time to savings by emptying (T_0)

$$0 = s_r W + (z - \bar{c})T_0$$

$$\Rightarrow T_0 = \frac{s_r W}{\bar{c} - z}$$

For token cost Feedback loop: Token Price Increase, more employees get fired. Another important factor to remember, that almost no employee goes back to its earlier job.

Here best case scenario in general assumed for a tech employee that he gets a nontech job after using all of his/her saving.

But the Wage Gap -

$$\Delta W = W_{Tech} - W_{Service} > 0$$

$$\Delta C_{direct} = -C_L \Delta W = -C_L (W_{Tech} - W_{Service}) = -C_L \Delta W$$

Reduction in private consumer demand

Displaced workers retooled across sectors

$$\Delta C_L \approx C_L \Delta Y_K \geq 0$$

[Assuming stays the same in the beginning of buffer]

National Economy:

$$Y_{national} = \underbrace{W_{Tech} L_{Tech} + W_{Service} L_S}_{Y_L} + \underbrace{Corporate + AI}_{Y_K}$$

$$Labor\ Share = \frac{W_{Tech} L_{Tech} + W_{Service} L_S}{Y_{national}}$$

$$Capital / Token\ share (S_K) = 1 - S_L = \frac{Y_{national} - Y_L}{Y_{national}}$$

$$1 = S_L + S_K$$

$$S_L = \frac{Y_L}{Y}, \quad S_K = \frac{Y_K}{Y}$$

labor share of national income

capital share by AI Corporate

Income based extreme storm

Consumption demand, $c = c_L Y_L + c_K Y_K$
 $= Y(c_L s_L + c_K s_K)$

consume for L

consume for corporate holders

$$c_L > c_K$$

Income redistribution: $\Delta Y_L = -L_{disp} (u_{tech} - u_{service}) < 0$

$$\Delta Y_K = \Delta Y - \Delta Y_L = \Delta Y + L_{disp} (u_{tech} - u_{service}) > 0$$

So, national income shifts away from labor, and concentrate on token revenue.

$$\Delta c = c_L \Delta Y_L + c_K \Delta Y_K$$

$$\Delta Y_K = \Delta Y - \Delta Y_L$$

$$\Rightarrow \Delta c = (c_L - c_K) \Delta Y + c_K \Delta Y$$

For consumer demand to shrink: ($\Delta c < 0$)

$$\frac{(c_L - c_K)}{c_K} \frac{\Delta Y}{|\Delta Y_L|} < \frac{c_L - c_K}{c_K}$$

Initial decline in consumer demand triggers a standard Keynesian multiplier effect across non-AI enterprise:

Final net change in aggregate macroeconomic demand $\leftarrow \Delta AD = \left(\frac{1}{1 - c_L} \right) \Delta c < 0$

Macroeconomic expenditure multiplier

downward economic spiral, further away from fixed points